

BART Ridership Analysis Report

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Dataset: BART_Rides (2021 Ridership Data)

Part 1: Feature Engineering

To analyze BART ridership patterns effectively, new features were engineered from the existing dataset:

- **Month Extraction:** The month was extracted from the date to facilitate monthly trend analysis.
 - **Day of the Week:** Each date was converted to a numeric day of the week, with Monday = 1 through Sunday = 7, allowing differentiation between weekday and weekend ridership.
 - **Line Identification:** The first letter of the **Lines** column was extracted to isolate the line code.
 - **Yellow Line Classification:** Stations were classified as belonging to the Yellow Line based on whether the first letter of their **Lines** entry was "Y."
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Part 2: Key Ridership Metrics

- **Total Ridership in 2021:**
The total number of passengers who rode BART in 2021 was **24,583,309**.
- **Weekday Ridership:**
A total of **20,281,852** passengers rode during weekdays, representing **83%** of all ridership.
- **Weekend Ridership:**
The remaining **17%** of riders used BART during weekends.
- **Yellow Line Ridership:**
Stations on the Yellow Line accounted for **19,425,399** passengers, approximately **79%** of total ridership.

Part 3: Data-Driven Recommendations

Weekday vs Weekend Ridership

The ridership data suggests that BART predominantly serves commuters traveling for work or school, with 83% of trips occurring on weekdays. To optimize service efficiency:

- **Increase service frequency during weekday peak hours** to accommodate high demand.
- **Moderate reduction of weekend service frequency** could optimize operational costs while maintaining reliable access.
- **Promote weekend ridership** through partnerships with local events and targeted marketing to increase off-peak usage.

Yellow Line Importance and Future Planning

Given that nearly four out of five riders use the Yellow Line, further data collection is recommended to fully assess its operational impact:

- Station-level boarding and alighting counts
- Passenger crowding and capacity data
- Delay and reliability statistics
- Transfer rates with other lines
- Passenger satisfaction surveys

This information would guide decisions regarding expanding, reducing, or enhancing Yellow Line services to maintain efficiency and rider satisfaction.

LevelUp Analysis: Downtown Ridership

- **Downtown Stations:** EM, MT, PL, and CC
- **Total Downtown Passengers:** 6,932,181 (28.2% of total ridership)

The downtown core stations handle over one-quarter of all BART riders, highlighting their critical role as hubs for work, tourism, and transit connections.

Recommendations:

- Prioritize frequent and reliable service to these stations during peak hours.
 - Increase staffing and maintenance efforts to manage high traffic.
 - Consider infrastructure upgrades such as platform expansions to accommodate growing demand.
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Conclusion

The 2021 BART ridership data reveals a commuter-focused system with significant concentration on the Yellow Line and downtown San Francisco stations. Strategic adjustments to service frequency, enhanced data collection, and targeted infrastructure investments can improve operational efficiency and rider experience across the network.
